High-Level Design Report

1. Introduction

1.1 Purpose of the system

The primary purpose of this system design is to bridge the gap between passive traffic surveillance and active forensic incident analysis. Traditional monitoring systems rely heavily on continuous human supervision, which is prone to fatigue and delayed response. This system aims to provide a fully autonomous, end-to-end architectural framework that can:

Automate Detection: Transition from raw pixels to structured incident data without human intervention.

Optimize Resource Allocation: Use a multi-stage "triggered analysis" strategy to ensure that computationally expensive models (like accident classification and LPR) only run when an anomaly is verified, making it ideal for edge-computing constraints.

Enhance Forensic Integrity: Generate objective, data-driven fault reports that can be used by law enforcement and insurance providers, thereby reducing ambiguity in traffic accident investigations.

In essence, the system is designed to transform existing camera infrastructures into intelligent diagnostic nodes capable of real-time decision-making and precise identification of traffic violations.

1.2 Design goals

The architecture of the system is governed by several critical design goals to ensure it is both practically deployable and technically superior:

Efficiency and Low Latency: Since the system is designed for edge deployment, it must maintain a high throughput. The primary goal is to process incoming frames in real-time, ensuring that the transition from anomaly detection to fault identification occurs in under 5 seconds.

Modularity and Decomposability: The system must be divided into independent subsystems (Tracking, Anomaly, Classification, LPR). This allows different modules to be updated or replaced (e.g., upgrading to a new YOLO version) without affecting the entire pipeline.

Robustness under Variable Conditions: The design must ensure consistent performance across diverse environmental states, such as night-time lighting, heavy rain, or glare, using robust feature extraction.

High Reliability (Pd vs. Fa): A paramount goal is to maximize the Probability of Detection (Pd) while maintaining a near-zero False Alarm (Fa) rate. The system should only trigger expensive diagnostic modules when there is high confidence in an anomaly.

Data Integrity and Accountability: Every generated report must be traceable and secure. The system must ensure that the video evidence and extracted license plate data are stored in a tamper-proof manner to serve as valid forensic evidence.

1.3 Definitions, acronyms, and abbreviations

Edge AI Node: A localized computing unit that performs artificial intelligence tasks at the data source (near the camera) to reduce latency.

Triggered Pipeline: An architectural pattern where high-compute modules remain dormant until a specific "trigger" event (anomaly) is detected.

RBAC (Role-Based Access Control): A security method that restricts system access based on the roles of individual users within an organization.

Spatio-Temporal Analysis: The study of data that changes across both space (pixel coordinates) and time (frame sequences).

NPU (Neural Processing Unit): A specialized hardware accelerator designed to accelerate machine learning algorithms.

Asynchronous Flow: A design pattern where a process operates independently of the main program flow, preventing bottlenecks during heavy computation.

1.4 Overview

The Anomaly-Based Accident Detection and Faulty Vehicle Identification System is engineered as a high-performance, autonomous forensic tool. The proposed architecture transitions from a continuous Perception Layer to a reactive Diagnostic Engine, ensuring that the system remains computationally efficient on edge hardware.

The design is decomposed into four primary subsystems that handle everything from raw video ingestion to encrypted report generation. By utilizing an event-driven control flow and a hybrid storage strategy, the system provides a robust solution for real-time traffic monitoring. Special emphasis is placed on security and data integrity, ensuring that every detected accident is documented with forensic-grade evidence including 10-second video clips and OCR-verified license plate data. This high-level design serves as the blueprint for a scalable, reliable, and intelligent traffic safety infrastructure.

2. Current software architecture

3. Proposed software architecture

3.1 Overview

The proposed software architecture is a modular, event-driven pipeline optimized for real-time edge intelligence. Unlike monolithic architectures, our design separates high-frequency data ingestion from low-frequency, high-compute diagnostic tasks. The architecture follows a "Triggered Pipeline" pattern:

Ingestion Layer: Captures raw video streams and converts them into normalized frames.

Processing Layer: Performs lightweight object tracking and continuous anomaly scoring. This layer runs indefinitely to maintain situational awareness.

Diagnostic Layer: Activated only when an anomaly is verified. It executes deep learning models for classification and forensic identification.

Reporting Layer: Aggregates findings and pushes them to a persistent database or a notification service.

By decoupling the continuous monitoring from the intensive analysis, the architecture ensures that the system remains responsive even under heavy traffic loads, preventing frame drops and ensuring high reliability.

3.2 Subsystem decomposition

Perception and Tracking Subsystem

This subsystem is the entry point of the pipeline. It handles high-speed video decoding and utilizes a lightweight object detector (e.g., YOLO-based) paired with a tracking algorithm (e.g., ByteTrack). Its primary output is a serialized stream of "Tracklets," which represent the position, velocity, and unique ID of every vehicle in the scene.

Anomaly Analysis and Triggering Subsystem

Acting as the system's "gatekeeper," this subsystem evaluates the motion vectors provided by the tracking module. It uses a spatio-temporal model to calculate an anomaly score. If the score exceeds the predefined confidence threshold (indicating a collision or illegal maneuver), it triggers the high-compute diagnostic models and sends a signal to the buffer manager to lock the relevant video segment.

Diagnostic and Forensic Identification Subsystem

Once triggered, this subsystem activates two parallel branches:

Event Classifier: Analyzes the 10-second clip to determine the accident category (Rear-end, Side-impact, etc.).

Identity Extractor (LPR/OCR): Focuses on the primary faulty actor to extract license plate details. This subsystem is designed to be dormant during normal traffic flow to conserve hardware resources.

Data Management and Reporting Subsystem

The final subsystem is responsible for synthesizing the metadata from all previous modules. It formats the results into a human-readable forensic report, handles database persistence, and manages the access control policies for stored evidence.

3.3 Hardware/software mapping

Platform Selection and Justification

The system is designed to be deployed on a High-Performance Edge AI Computing Platform. This selection is based on the requirement to process multi-stream video data locally without relying on cloud infrastructure, ensuring low latency and data privacy. As illustrated in **Figure 1**, the architecture establishes a direct connection between the sensing units and the edge computing node to minimize data transit times.

Hardware Requirements: The target platform must feature a multi-core CPU for general logic and a dedicated Hardware Accelerator (GPU or NPU) with support for parallel computing (e.g., CUDA or OpenCL) to execute deep learning models in real-time.

Operating System: A Linux-based OS (e.g., Ubuntu LTS) is selected as the primary environment due to its extensive support for containerization (Docker), hardware drivers, and high-performance inference libraries.

Mapping of Subsystems to Hardware

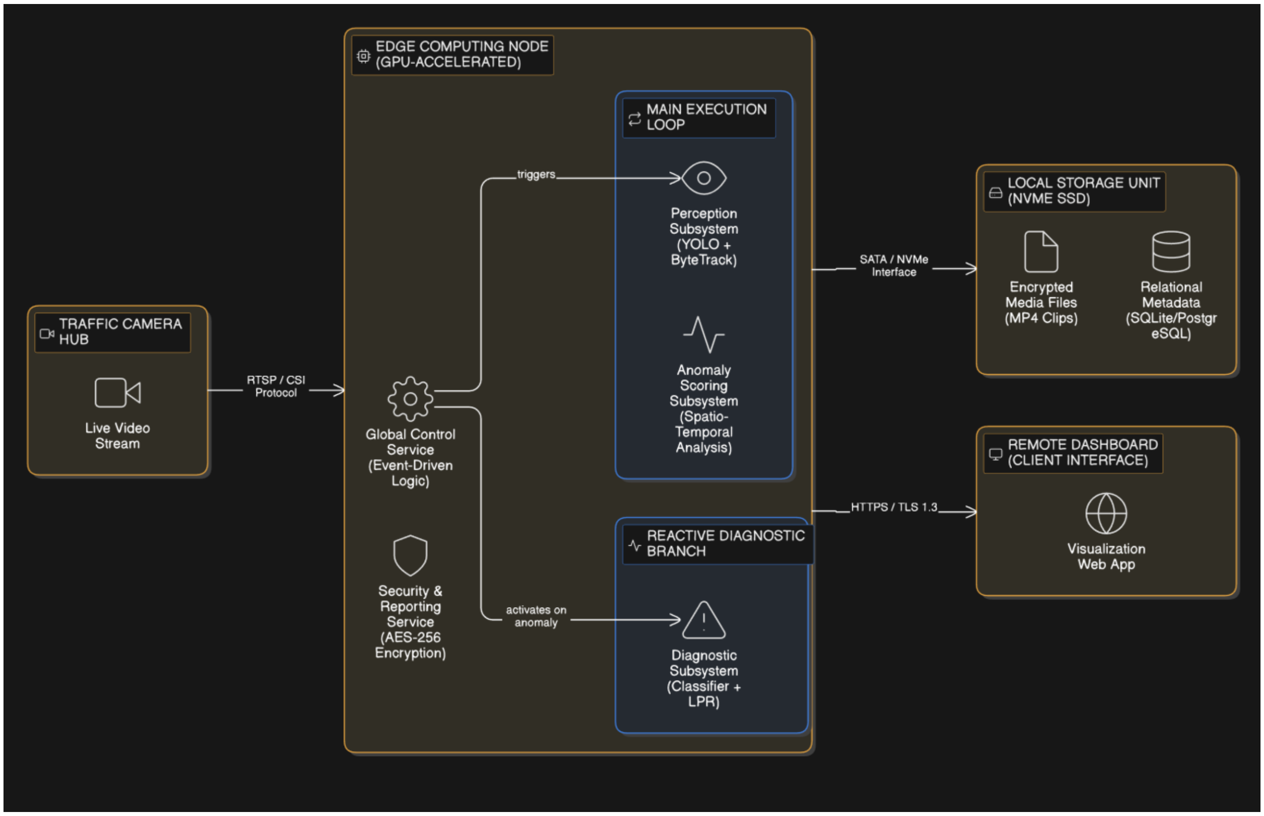
The mapping strategy ensures optimal resource distribution across the chosen edge hardware, partitioning tasks between continuous monitoring and reactive analysis as visualized in the UML Deployment Diagram:

Perception & Tracking: Mapped to the Hardware Accelerator (GPU/NPU). This module requires high throughput to maintain real-time frame rates during object detection. The diagram reflects this as the "Main Execution Loop," which remains active to process incoming RTSP streams.

Anomaly Detection: Primarily mapped to the CPU, utilizing optimized mathematical libraries. When an anomaly is suspected, the GPU assists in calculating complex spatio-temporal scores.

Diagnostic (Classification & LPR): Mapped to the Hardware Accelerator's Memory. These high-compute models are managed dynamically, being fully activated only upon a trigger signal to optimize power consumption and thermal performance. This "Reactive Diagnostic Branch" is only engaged when the Global Control Service issues an activation signal.

Data Management: Mapped to High-Speed Local Storage (e.g., NVMe SSD) to ensure that 10-second video clips and encrypted logs are written without causing bottlenecks in the pipeline. The "Local Storage Unit" node in Figure 1 demonstrates the hybrid storage approach for both relational metadata and encrypted media.



[Figure 1: Hardware/Software Mapping Deployment Diagram]

The deployment diagram showcases the physical and logical mapping of the system. It highlights the event-driven control flow where the Global Control Service acts as the orchestrator, ensuring that high-compute diagnostic artifacts are only deployed to the GPU memory upon a verified anomaly trigger, thereby adhering to our design goals of computational efficiency and low latency.

3.4 Persistent data management

Data Storage Strategy

The system adopts a hybrid storage strategy to balance high-speed access with structured data integrity. Since the system operates at the edge, data management must be lightweight yet robust:

Relational Metadata (SQLite/PostgreSQL): A lightweight relational database is used to store structured incident data, including timestamps, vehicle IDs, accident types, and detected license plates. This ensures fast querying and filtering of historical records.

Unstructured Multimedia (File System): 10-second accident clips (MP4/AVI) and high-resolution plate snapshots are stored directly on the local file system. The database maintains the file paths as foreign keys to link metadata with visual evidence.

Short-term Buffer (RAM/Cache): A circular buffer in memory stores the last 15-20 seconds of the live stream to allow for "pre-event" video extraction when an anomaly is triggered.

Data Security and Lifecycle

Encryption: All stored media and sensitive plate data are encrypted to prevent unauthorized access and ensure forensic validity.

Retention Policy: To prevent storage overflow on edge devices, an automated cleanup policy is implemented. Data is categorized as "Critical" (confirmed accidents) or "Routine" (unconfirmed anomalies). Critical data is kept for 30 days or until manually offloaded, while routine logs are overwritten every 48 hours.

3.5 Access control and security

Access Control Policy

The system employs a Role-Based Access Control (RBAC) model to ensure that sensitive data is only accessible to authorized personnel. Access levels are strictly partitioned:

Administrator: Full access to system configurations, hardware logs, and security settings. Can manage user accounts but cannot modify recorded forensic evidence.

Operator (Traffic Controller): Can view live streams, monitor real-time anomaly alerts, and export incident reports. Access to historical archives is limited to a "read-only" basis.

Auditor / Law Enforcement: Has specialized access to verified accident reports and encrypted video clips for legal investigations.

System (Automated): Internal modules have "write-only" access to the storage during the reporting phase to ensure that automated processes do not interfere with existing records.

Security Mechanisms

To protect the integrity of the edge node and the data it generates, the following security layers are implemented:

Data-at-Rest Encryption: As specified in the data management section, all incident-related files are encrypted.

Secure Communication: Any data transmission between the edge device and external dashboards or cloud storage is performed over HTTPS/TLS 1.3 to prevent man-in-the-middle attacks.

Integrity Verification (Hashing): Each generated report and video clip is assigned a digital fingerprint at the moment of creation. This allows the system to detect if any forensic evidence has been tampered with or corrupted.

Physical Security Ports: Since the device is deployed in the field, unused I/O ports are software-disabled, and an encrypted boot process (Secure Boot) ensures that only authorized firmware can run on the hardware.

3.6 Global software control

Control Flow Strategy

The system utilizes an event-driven, asynchronous control flow to maintain real-time performance. The execution logic is divided into a continuous main loop and a reactive diagnostic branch:

The Main Loop (Execution State: Monitoring): The central controller continuously polls the Ingestion Layer for new frames. It orchestrates the flow between the Perception (Tracking) and Anomaly modules. This loop runs at a high frequency (e.g., 30 FPS) to ensure no critical data is missed.

The Reactive Branch (Execution State: Diagnostic): When the Anomaly Subsystem signals a "Confidence Threshold Breach," the global controller interrupts the routine flow (or spawns a parallel thread). It triggers the Diagnostic Subsystem, providing it with the specific temporal window (video clip) from the buffer.

Termination and Recovery: The controller monitors the health of each subsystem. If a module fails (e.g., a GPU driver error), the global control unit initiates a "Safe Restart" sequence to restore the monitoring state without losing persistent data.

Synchronization and Concurrency

To optimize the target edge hardware, the system employs Multi-threading and Producer-Consumer patterns:

Producer: The Ingestion and Tracking units act as producers, filling a shared circular buffer with processed metadata and frames.

Consumer: The Anomaly and Diagnostic units act as consumers, pulling data from the buffer only when computational resources are available or when an event is triggered.

Global State Machine: A state machine manages transitions between Idle (No Traffic), Monitoring (Active Flow), Alert (Anomaly Suspected), and Reporting (Evidence Compilation) states.

3.7 Boundary conditions

Hardware and Connectivity Failures

The system is designed to handle hardware-level disruptions to ensure continuous monitoring:

Power Instability: In the event of a power fluctuation or sudden shutdown, the system utilizes an atomic commit strategy for its database. This ensures that the metadata of the last processed incident is not corrupted upon reboot.

Camera Disconnection: If the video feed is lost (due to physical damage or cable failure), the global controller enters a "Retry-and-Alert" state. It attempts to re-establish the stream while simultaneously sending a critical heartbeat failure notification to the central dashboard.

Storage Saturation: When the local storage reaches 90% capacity, the system triggers its automated retention policy (First-In-First-Out for non-critical data) to ensure that the recording of new incidents is never interrupted.

Environmental and Operational Extremes

Visual Obstruction: In cases of extreme weather (heavy fog, blizzard) or lens occlusion (dirt/vandalism) that leads to very low confidence scores in detection, the system flags the frames as "Poor Visibility." It may switch to a more robust, lower-resolution processing mode to maintain basic motion tracking.

Maximum Load: If an unusual number of anomalies occur simultaneously, the system prioritizes the "Buffer Lock" (saving raw data) over "Real-time Classification." This ensures that evidence is preserved even if the processing of the reports is slightly delayed due to computational spikes.

High-Speed Incidents: To handle extremely fast collisions, the system maintains a high-frequency temporal buffer, allowing it to "look back" in time and reconstruct events that occurred within milliseconds.

4. Subsystem services

Perception and Tracking Services

This subsystem provides the foundational data layer for the entire pipeline.

Object Detection Service: Provides real-time coordinates and class labels (car, truck, motorcycle) for all detected entities.

ID Management Service: Maintains temporal consistency by assigning and providing persistent unique identifiers to vehicles across consecutive frames.

Trajectory Stream Service: Exports a continuous stream of (x, y) coordinates and velocity vectors for each active ID.

Anomaly and Triggering Services

This subsystem acts as the logic engine that determines when high-compute resources should be activated.

Scoring Service: Continuously evaluates motion patterns and provides a numerical "Anomaly Score" based on spatio-temporal deviations.

Event Trigger Service: Broadcasts an activation signal to the Diagnostic subsystem when the score exceeds the confidence threshold.Buffer

Lock Service: Commands the data management layer to persist a specific temporal window (e.g., (T-5) to (T+5) seconds) upon a detected incident.

Diagnostic and Identification Services

These services are reactive and perform deep analytical tasks.

Classification Service: Analyzes locked video segments to provide a categorical description of the accident (e.g., "Rear-end Collision").

Actor Attribution Service: Identifies which vehicle ID is the primary cause of the anomaly through proximity and vector analysis.

Identity Extraction (LPR/OCR) Service: Provides the alphanumeric license plate string and a high-resolution snapshot of the faulty vehicle.

Management and Reporting Services

The interface layer for external users and persistent storage.

Report Generation Service: Synthesizes data from all subsystems into a finalized, encrypted forensic document.

Query & Archive Service: Provides an interface for searching historical incidents based on date, location, or plate number.

Health Monitoring Service: Monitors the status of GPU/CPU utilization and camera connectivity, providing real-time system diagnostic data.

5. Glossary

Atomic Commit: A database operation strategy ensuring that a transaction is either fully completed or not at all, preventing data corruption during power failures.

Causality-Based Fault Attribution: The analytical process of determining which vehicle's actions directly led to a collision by evaluating their motion vectors prior to the incident.

Circular Buffer: A data structure that uses a single, fixed-size buffer as if it were connected end-to-end; used here to temporarily hold recent video frames for "pre-event" extraction.

Decoupling: An architectural principle of separating system components so they can operate and be updated independently, reducing dependencies between modules.

Event-Driven Architecture: A software design pattern where the flow of the program is determined by events such as sensor outputs or specific detection triggers (e.g., an accident).

Hybrid Storage: A management strategy that combines different storage methods, such as using a relational database for text data and a standard file system for video files.

Persistence: The characteristic of data that outlives the process that created it, achieved through saving information to non-volatile storage like an SSD.

Reactiveness: The ability of the system to remain dormant under normal conditions and activate specific high-compute resources only when a specific stimulus is detected.

Subsystem Decomposition: The process of dividing a complex system into smaller, more manageable, and specialized functional units.

Tamper-Proof: A design feature that ensures data cannot be modified or deleted by unauthorized users, maintaining the integrity of forensic evidence.

Temporal Consistency: Ensuring that an object’s identity and properties remain logically linked and stable as it moves across consecutive frames in a video.

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